

SPECIFICATION
Preliminary

WHEC
WEIHAI HUALING OPTO-ELECTRONICS CO., LTD

Notice: This is not a final specification.
Some parameters are subject to change

COLOR CONTACT IMAGE SENSOR

LC3R216N-8008

Approved by customer		

WHEC WEIHAI HUALING OPTO-ELECTRONICS CO.,LTD.
No.179, Keji Road, Hi-Tech. IDZ Weihai Shandong, China.

Tel: +86-631-569-8128 (Overseas dep.)
+86-631-569-8618 (Domestic dep.)
Fax: +86-631-568-8608
E-mail: sales@whec8.cn

REVISION					<u>Approved</u>
Rev	Description	Date	Approved	Designed	
A	---	2023.12.18	Liuzhenxiang	Duanping	
					<u>Checked</u>
					<u>Designed</u>

1. Description

This specification is applied to LC3R216N- 8008 Color Contact Image Sensor module .

2. Scope

This LC3R216N-8008 is a Color CIS consists of a Rod Lens Array, a color LED light source and an array of linear MOS image sensor.

3. Outline

Item		Specification		Note
Scanning width		216 mm		
Sensor element density		600DPI(WHEC TEST)	300DPI	300DPI:CNT=GND 600DPI:CNT=VDD
Effective number of sensor elements		5,104 elements 41 [#] to 5,144 [#] (Full 5,184elements)	2,552 elements 21 [#] to 2,572 [#] (Full 2,592elements)	
scanning speed	Color(R/G/B)	228×3μsec/line	120×3μsec/line	
	Black&white	228μsec/line	120μsec/line	
Clock speed		8.0 MHz		
Rod lens array		Single row		L22
Light source		Red λp = 630nm ± 15nm 60mA Green λp = 520nm ± 15nm 60mA Blue λp = 465nm ± 10nm 60mA		
Power supply		+3.3V x 100 mA		
Data output 3 analog output		Block#1 1728 pixels Block#2 1728 pixels Block#3 1728 pixels	Block#1 864 pixels Block#2 864 pixels Block#3 864 pixels	Synchronous
Dimensions		Figure 1		

Note: Clock Speed f must satisfy the following status:

$$f > (n + 92) / \text{tint}$$

f : Clock speed

n: Full sensor elements number of every block.

n=864 300dpi

n=1728 600dpi

tint: Scanning speed

4. Image Data Output Characteristics (Ta = 25°C)

The shipment test in WHEC is done on the condition of this table.

In colour mode

Item	Symbol	Specification			Note
		Red	Green	Blue	
DC supply voltage	VDD	+3.3V			Detector, Logic
LED supply voltage	VLED	<3.0V	<5.0V	<5.0V	
LED supply current	ILED	60mA	60mA	60mA	
White image target		0.05~0.09 OD			
Video reference	Vref	800 ± 200 mV			4.1
Dark output minimum	Vdmin	≥ -150mV			4.2
Dark output maximum	Vdmax	≤ +150 mV			4.3
White output maximum	Vpmax	300 ± 100 mV T.B.D			4.4
White output uniformity	UEp	Less than 55%			4.5
MTF (300dpi)		>30%	>30%	>30%	4.6 71.37 lppi
Linearity	Gamma	1.0 ± 0.05			
Linearity uniformity	LU	Less than 6%			4.7

B&W Mode

Item	Symbol	Specification			Note
		Red	Green	Blue	
DC supply voltage	VDD	+3.3V			Detector, Logic
LED supply voltage	VLED	<3.0V	0V	0V	
LED supply current	ILED	60mA	-	-	
White image target		0.05~0.09 OD			
Video reference	Vref	800 ± 200 mV			4.1
Dark output minimum	Vdmin	≥ -150mV			4.2
Dark output maximum	Vdmax	≤ +150 mV			4.3
White output maximum	Vpmax	300 ± 100 mV T.B.D			4.4
White output uniformity	UEp	Less than 55%			4.5
MTF(300dpi)		>30%			4.6 71.37 lppi
Linearity	Gamma	1.0 ± 0.05			
Linearity uniformity	LU	Less than 6%			4.7

The output level of image signal like white and dark and MTF is defined at the point of “ts” which described in section 6. A test target is set on the reading position as outlined in Figure 1.

4.1 Vref

Video reference voltage.

Vref is outputted from connector pin #7. Vdmim and Vdmax are based on Vref.

4.2 Vdmin

As shown in Figure 2, Vdmin is the minimum in the dark output signal (turning off the LED). Every other parameter is defined by Vdmin as a reference.

4.3 Vdmax

As shown in Figure 2, Vdmax is the maximum in the dark output signal (turning off the LED).

4.4 Vpmax

As shown in Figure 2, Vpmax is the maximum white output signal and is defined by:

$$Vpmax = \text{MAX} [Vp(n)]$$

Vp(n) is the output signal of the nth pixel using a white image target.

4.5 UEp

UEp is the white output non-uniformity with dark signal subtracted and is defined by:

$$UEp = ((VEpmax - VEpmin) / (VEpmax)) \times 100\%$$

VEpmax = MAX [VEp(n)]; is the maximum effective output signal

VEpmin = MIN [VEp(n)]; is the minimum effective output signal

VEp(n) is the effective output signal of every pixel and is defined by:

$$VEp(n) = Vp(n) - Vd(n)$$

4.6 MTF

MTF is defined by:

$$MTF = \text{MIN} \{[(Vmax - Vmin) / VEp]\} \times 100\%$$

Vmax is the maximum output signal using the MTF image target

Vmin is the minimum output signal using the MTF image target

VEp is the effective output signal.

4.7 Linearity Uniformity

LUg is measured following procedure and defined;

Step1. Test Target

The white image target is used as a test target. This target must not be moved while this test is being operated.

Step2. LED adjustment

Tred, Tgrn and Tblu should be adjusted according to Figure 8 procedure.

Step3. Dark and White correction

Dark and White correction must be done for every each pixel.

Step4. LED on time set

Tred, Tgrn and Tblu should be changed as following;

Tred/2, Tgrn/2, Tblu/2.

Step5. Compute LUG

LUG should be computed for each color as;

$$LUG = \sqrt{D_{gave} - D_{gextm}}$$

Dgave is the average of Vg (n). Vg (n) should be got more than 8 times sampling.

Step6. LED on time set

Tred, Tgrn and Tblu should be changed as followed and compute LUG regarding to Step5;

Tred/4, Tgrn/4, Tblu/4.

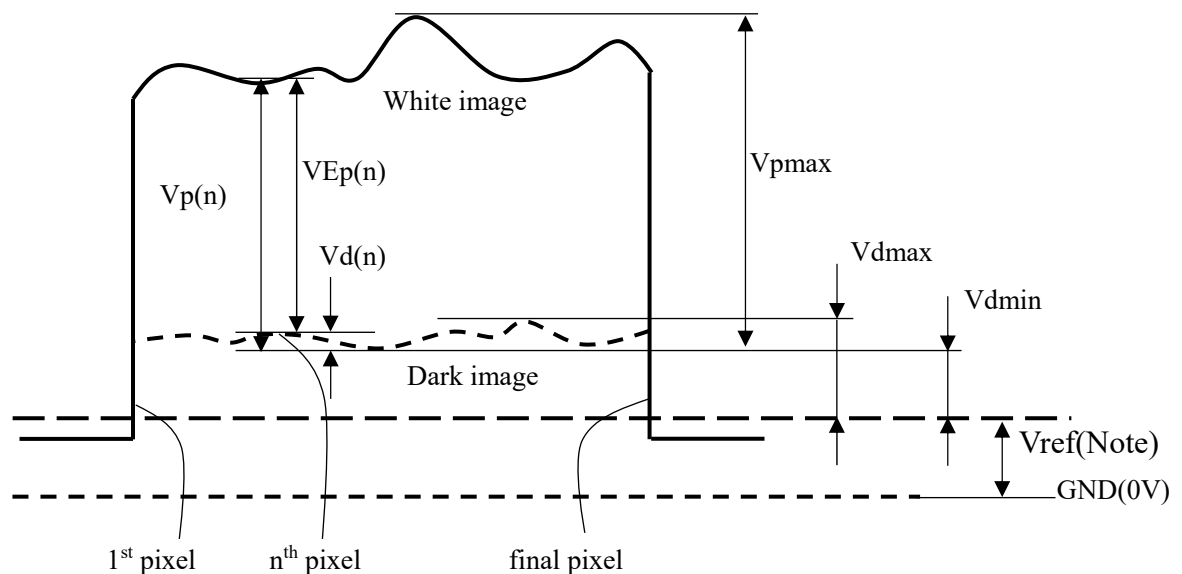
Step7. LED on time set

Tred, Tgrn and Tblue should be changed as followed and compute LUG regarding to Step5;

Tred/8, Tgrn/8, Tblu/8.

4.8 Correction of Dark and White uniformity

For the best performance two points correction (dark and white) is strongly recommended.



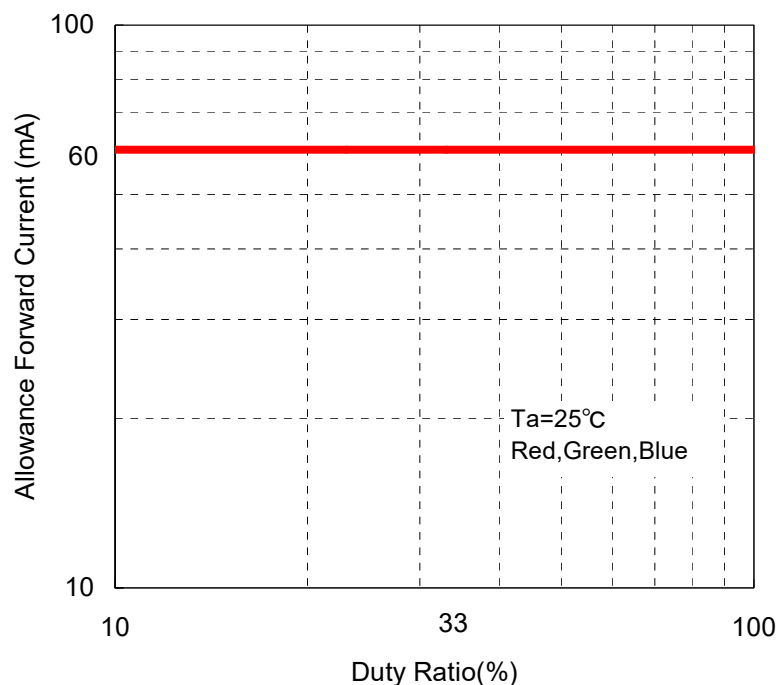
Note: Vref is the reference voltage for video signals. It can be used as the reference voltage. Do not use the GND instead of Vref.

5.Maximum Rating

Item	Symbol	Specification	Note
DC supply voltage	VDD	+3.3V $\pm$ 0.17V	
Input voltage	VIN	0 ~ VDD+0.3V	SI, CLK
Ambient temperature	Ta	0 ~ +50 °C	Operating
		-20 ~ +60 °C	Non-operating
Ambient humidity		10 ~ 90%RH	Avoid a build up condensation
Maximum operating Temperature		65 °C 30minuts MAX	

LED

Parameter	Symbol	Red	Green	Blue	Notes
DC Forward Current	IF	60 mA	60 mA	60 mA	
Pulse Forward Current	IFP	60 mA	60 mA	60 mA	
DC Reverse Voltage	VR	3.0 V	5.0V	5.0V	

**Figure 3. Duty Ratio vs Allowance Forward Current**

6. Electrical Characteristics (Ta = 25 °C)

Item	Symbol	Condition	Specification			Unit
			Min.	Typ.	Max.	
DC Supply Voltage	VDD	GND reference	3.13	3.3	3.47	V
DC Supply Current	IDD	VDD = 3.3V		100	120	mA
LED Forward Voltage (Note1)	VFred	IF=20mA	1.8	2.0	2.1	V
		IF=30mA	1.9	2.0	2.2	V
		IF=40mA	2.0	2.1	2.2	V
		IF=60mA	2.0	2.15	2.3	V
	VFgrn	IF=20mA	2.8	3.0	3.1	V
		IF=30mA	2.9	3.0	3.1	V
		IF=40mA	3.0	3.1	3.2	V
		IF=60mA	3.2	3.3	3.5	V
	VFblu	IF=20mA	2.9	3.0	3.1	V
		IF=30mA	3.0	3.1	3.2	V
		IF=40mA	3.1	3.2	3.3	V
		IF=60mA	3.3	3.4	3.5	V
Input voltage (Note2)	VIH	SI,CLK	2.4			V
	VIL				0.5	V
Input Current (Note2)	IIH	SI,CLK			5	mA
	IIL		-0.5			μA
Clock frequency	f	CLK	7.5	8.0	8.5	MHz
Clock pulse duty		tw(T)/to; to=1/f	48	50	52	%
SI setup time	Tsu	SI-CLK	60		TW	ns
		SI-CNT	-10		10	
SI hold time	Th	SI-CLK	60		5×TW	ns
		SI-CNT	-10		10	
Data output stability time	Ts	CLK-SIG	20	30	40	ns

Note 1: LED Forward Voltage is a reference value, and it should be got by experiment in actual working condition.

Note 2: I/O transceiver is recommended for input signals to improve driving performance.

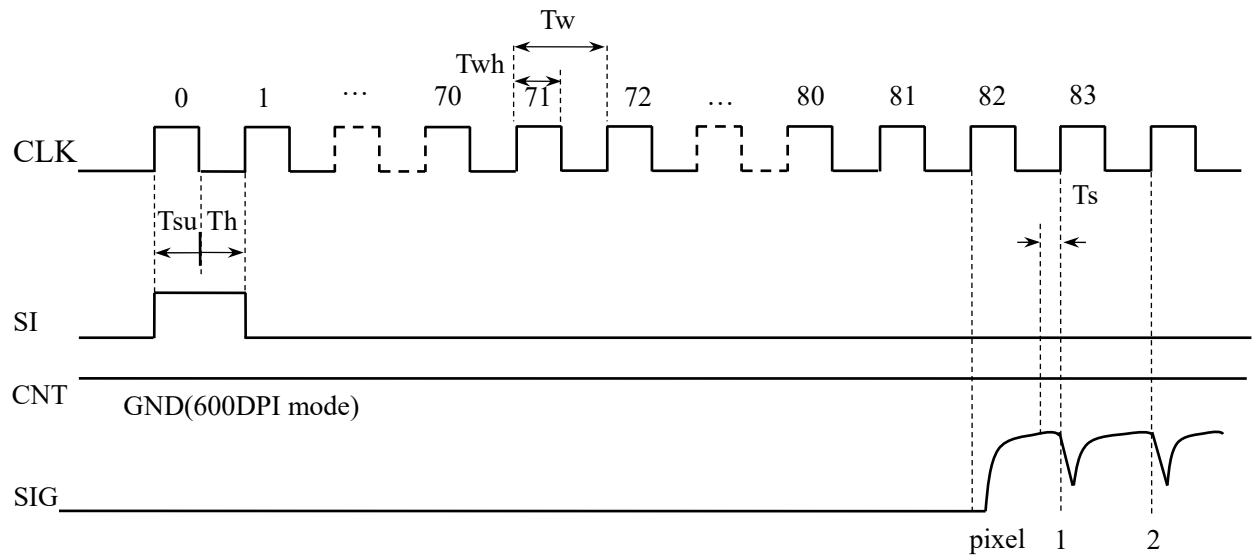


Figure 4. Timing Diagram

7. Reliability

The following table satisfies the reliability when the CIS is operated continuously under standard operating conditions as specified in section 4.

Item	Variable Amount (%)	Note
White output	Initial level +10% -20%	1000Hr
	Initial level +10% -30%	5000Hr

8. Precautions before use:**8.1 Polycarbonate sheet surface**

The Glass surface should be kept clean. Don't wipe the glass surface with hand. If the glass surface gets dirty, wipe the glass surface gently with a clean cloth soaked in alcohol. The glass surface should be wiped very carefully.

8.2 Extracting / Inserting the connector

The maximum number of times that the connector should be extracted and connected is 10. If the connector is inserted / extracted more than 10 times, the connector 'burrs' will be eroded, thereby making the connector ineffective.

8.3 Stable operation

(1) The connector pins should not be touched by bare hand or electrostatic charge materials.

(2) Noise

- a. Insert a low frequency noise suppressing capacitor (100uF) between VDD (+3.3V) and GND. A high frequency noise suppressing capacitor is already integrated into the circuit.
- b. Ensure that the sensor connecting cables are 30cm or less in length. The CLK and GND, SIG and GND and VLED and LEDr, LEDg, LEDb, respectively should form twisted cable pairs.

(3) Latch up

When the supply voltage is higher than the absolute maximum, latch up will cause the sensor to break, even if the voltage is caused by a surge. If the current varies rapidly in the external circuit, or when the power is turned on an off very frequently, ensure that the voltage of each terminal does not exceed the values indicated in below.

(4) LED circuit

As shown in Figure.5 LED circuit has not any resistance. Be careful not to connect the LED circuit to power supply directly without current limit resistors.

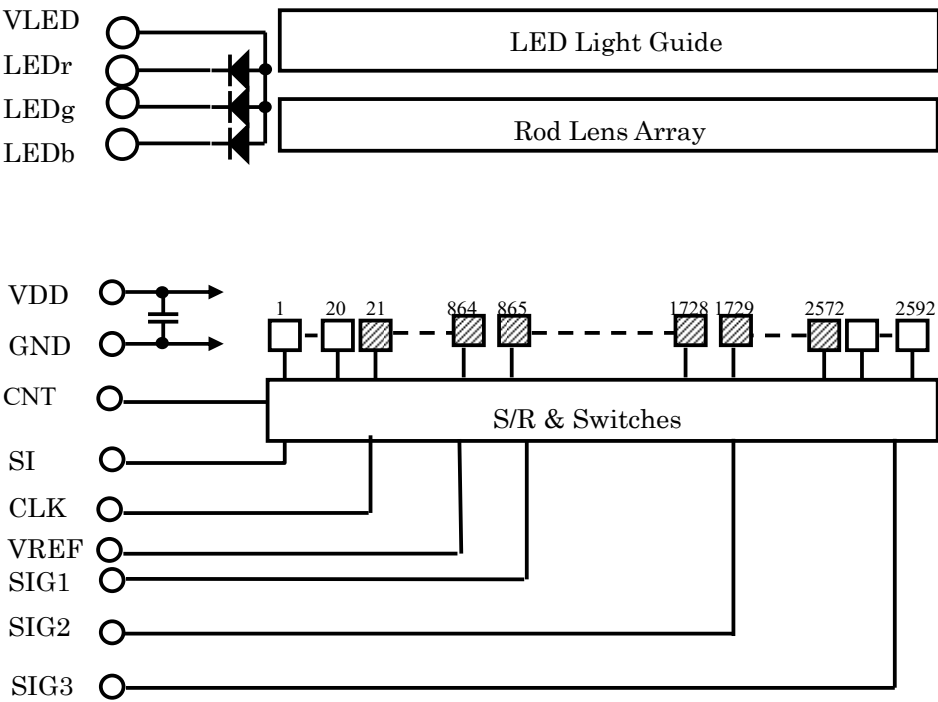
(5) Absolute maximum ratings

Item	Symbol	Condition	Specification		Unit
			Min	Max	
Supply Voltage	VDD	GND reference	-0.3	+4.5	V
Input voltage	Vin	SI,CLK	GND-0.3	VDD+0.3	V

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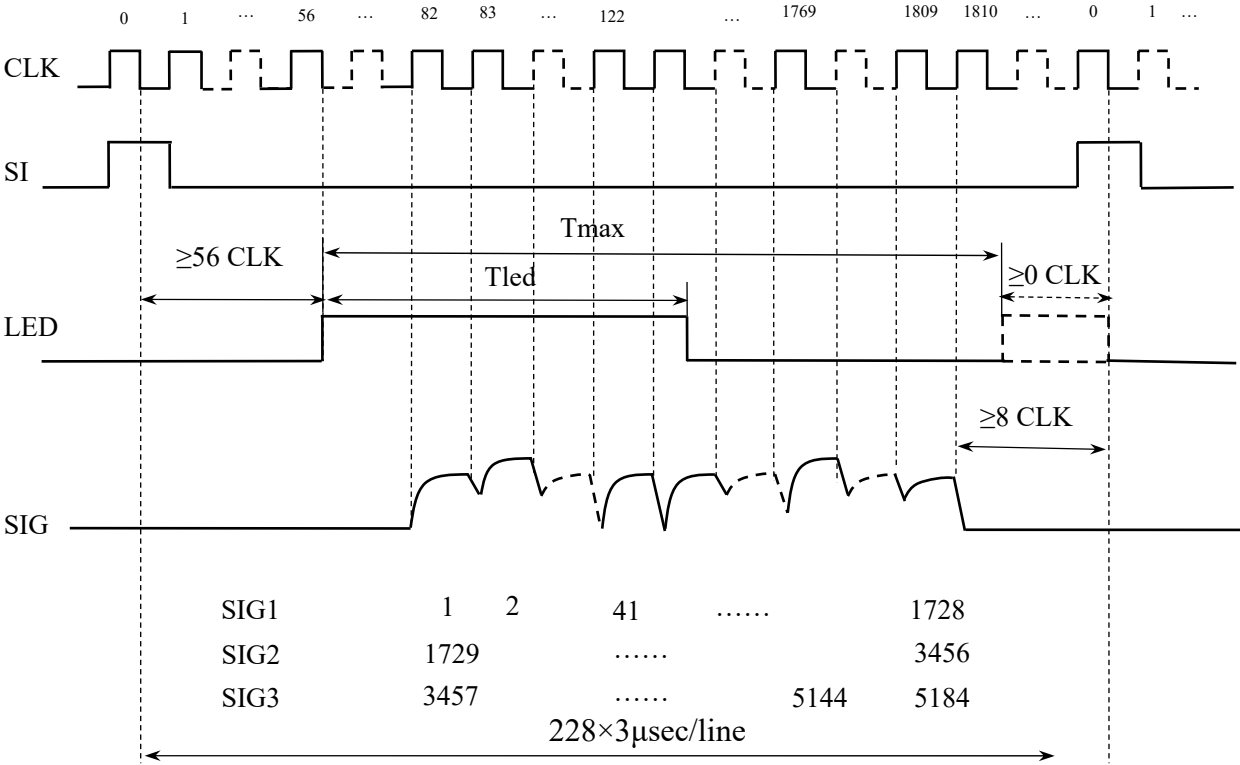
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Black marked detector element outputs are the actual image signal

Figure 5. Block Diagram



□ Color Mode

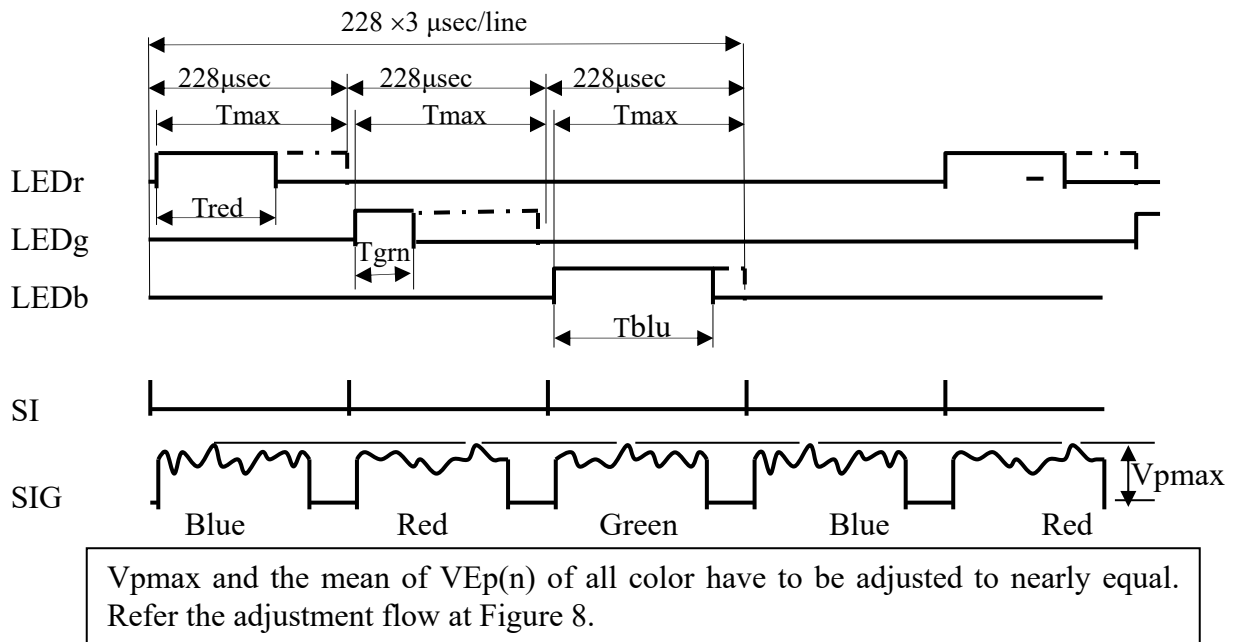


Figure 7. Color mode Timing Diagram(This is the WHEC shipping test condition)

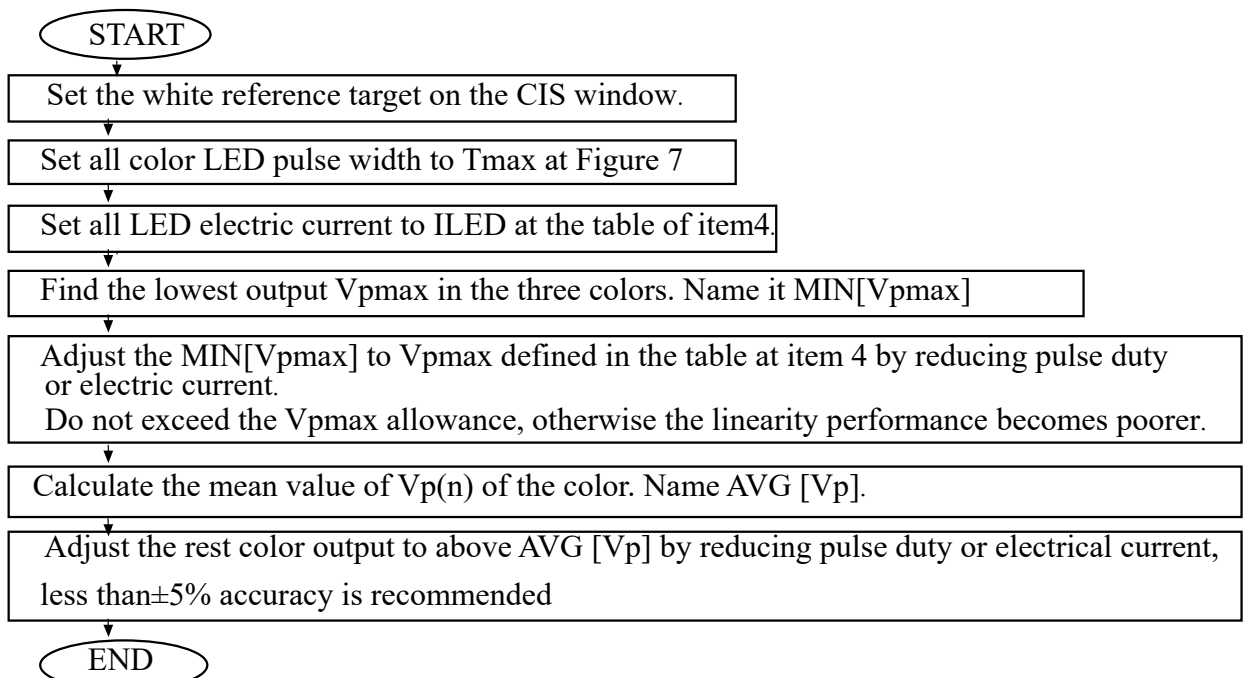
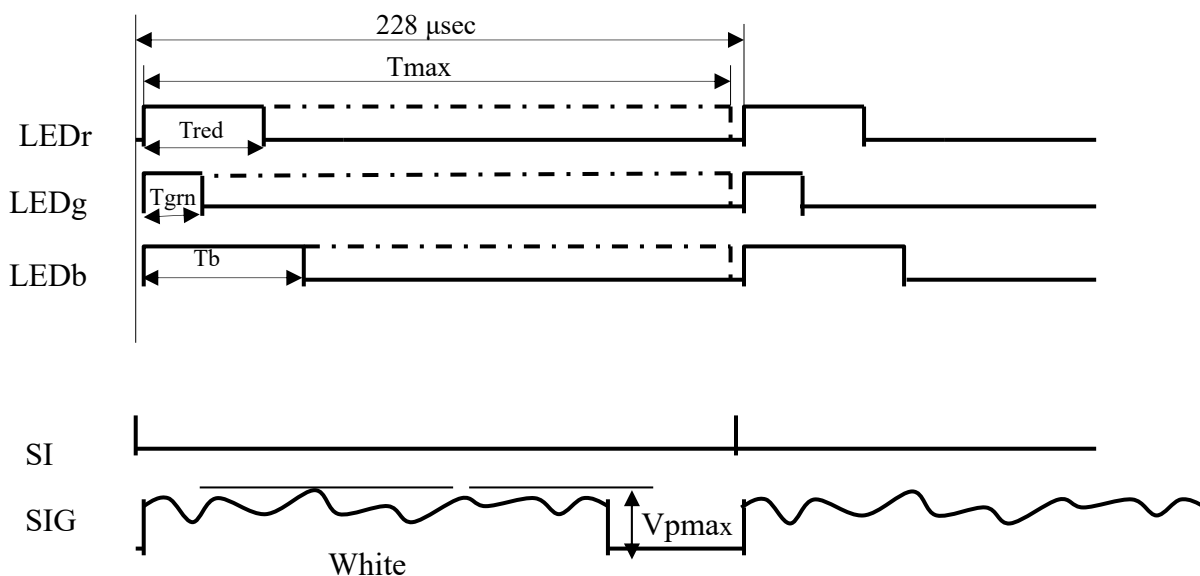


Figure 8. Flow Chart of Color mode Adjustment (This is the WHEC shipping test condition)

□ **B&W Mode White Light Source**



This is the sample for white light source application. Refer the adjustment flow chart at Figure 10.

Figure 9. B&W mode with White Light Source Timing Diagram

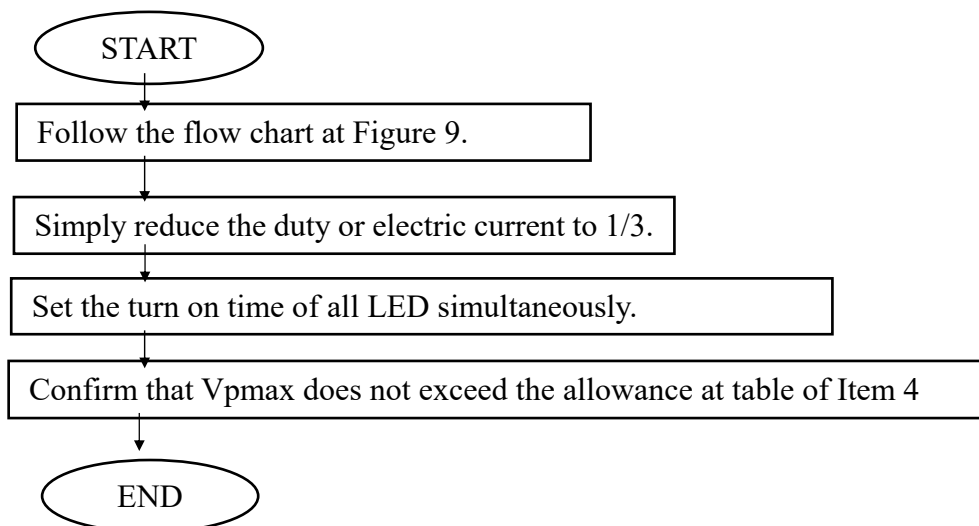
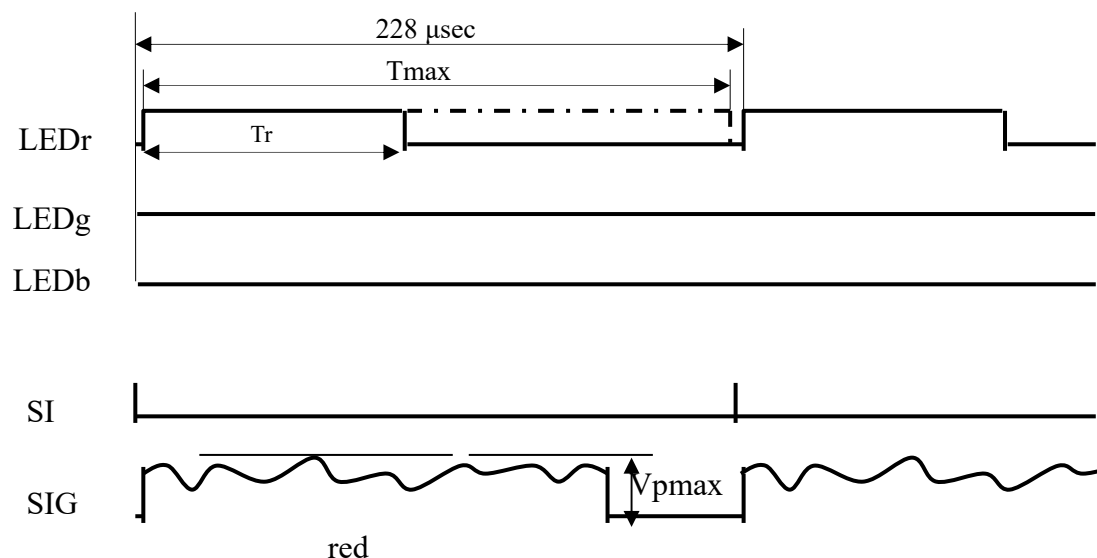


Figure 10. Flow chart of B&W mode with White Light Source Adjustmen

□ B&W Mode with Mono-Color Light Source



This is the example for White light source application. Refer the adjustment flow chart at Figure 12.

Figure 11. Mono-Color mode Light Source Timing Diagram

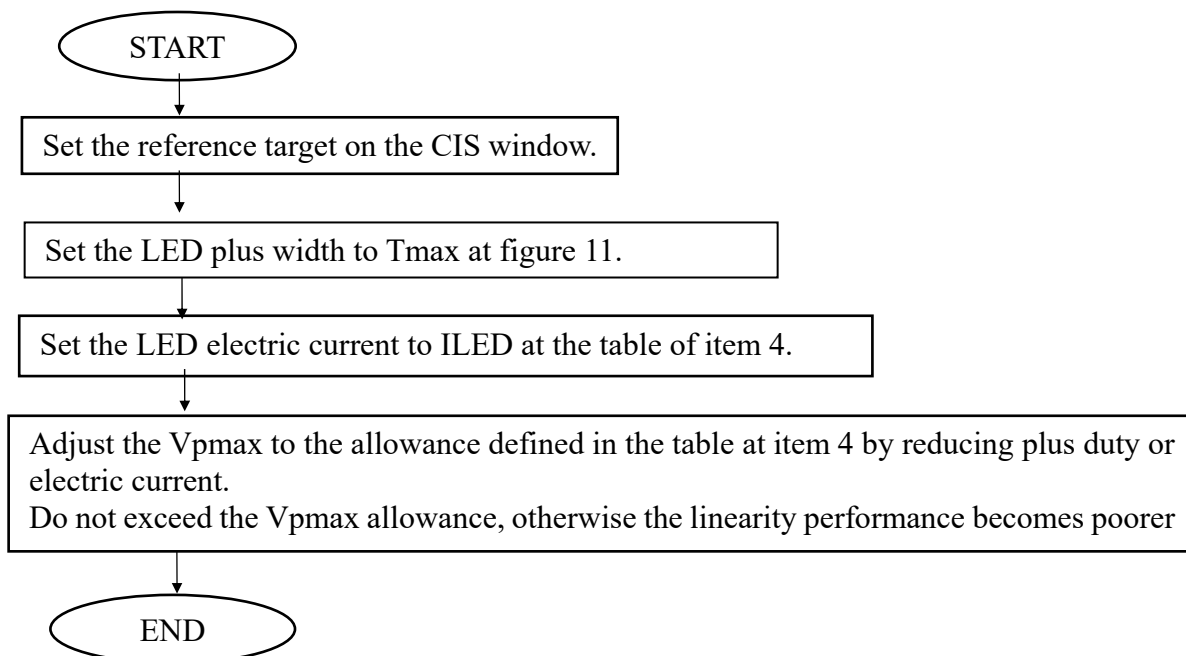


Figure 12. Flow Chart of Mono-Color Adjustment

Figure 1 Dimensions

